

SmartCart

Customer Segmentation System

Project Report — Unsupervised Machine Learning

Total Customers	Clusters Found	Silhouette Score	Algorithm
2,236	4	0.3793	Agglomerative

Dataset: 2,240 records | 22 features | Demographics, Spending, Engagement

1. Problem Statement

SmartCart is a growing e-commerce platform that was applying the same generic marketing strategy to all 2,240 customers — regardless of their spending behaviour, income, or engagement level. This resulted in wasted marketing budget, missed retention opportunities for high-value customers, and delayed identification of churn-prone users.

The goal was to build an intelligent customer segmentation system using unsupervised machine learning — grouping customers into meaningful clusters based on purchasing behaviour, engagement levels, and loyalty indicators — to enable data-driven, personalised marketing.

2. Dataset Overview

2.1 Raw Dataset

2,240 customer records across 22 attributes covering four domains:

Category	Features
Demographics	ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Dt_Customer
Spending	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds
Purchase Frequency	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth
Feedback	Recency, Complain, Response

2.2 Key Data Facts

- **Missing values:** 24 records in Income — filled with median (robust to the outlier at 666,666).
- **Outliers removed:** 3 customers with Age > 90, 1 with Income > 600,000. Final dataset: 2,236 records.
- **Average income:** 52,238 | **Average total spending:** 606 (std dev equals mean — highly skewed).
- **Wine dominates spending** at avg 304, followed by meat (167) and fish (38).
- **Campaign response rate:** 14.9% | **Complaint rate:** 0.9%.

3. Feature Engineering & Preprocessing

Six new features were derived from raw columns:

Feature	Derivation	Purpose
Age	2026 - Year_Birth	Interpretable age from birth year
Customer_Tenure_days	Days since enrollment (reference = max date)	Loyalty / longevity signal
Total_spending	Sum of all 6 Mnt* columns	Single unified spending metric

Total_Children	Kidhome + Teenhome	Household life stage proxy
Education (grouped)	5 levels collapsed to 3	Reduce cardinality and noise
Living_With	8 marital statuses collapsed to Together/Single	Fix data quality (YOLO, Absurd entries)

Categorical features (Education, Living_With) were one-hot encoded. All features were standardised with StandardScaler before PCA.

4. Exploratory Data Analysis

4.1 Pairplot — Key Features

Shows pairwise relationships and KDE distributions across income, recency, age, spending, and children. Income and spending show a clear positive correlation. Recency is uniformly distributed — it does not separate customers meaningfully.

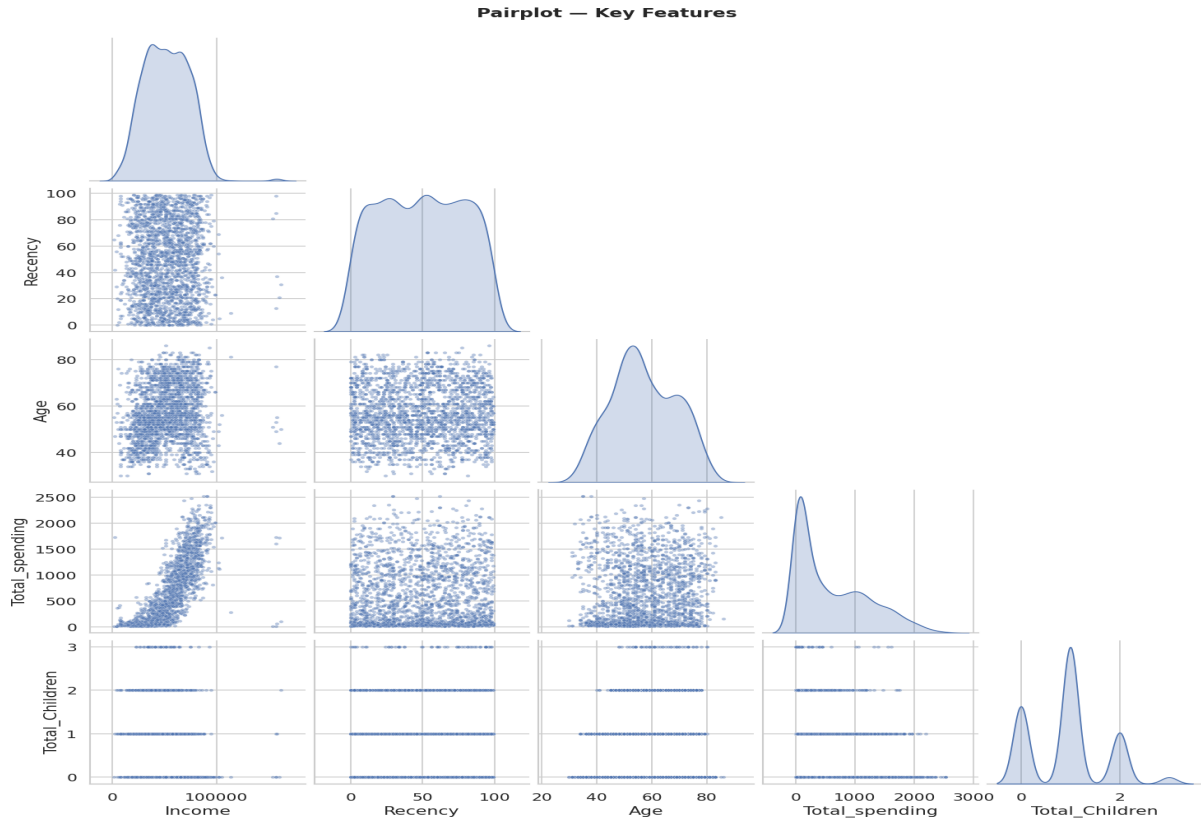


Figure 1 — Pairplot of key numerical features

4.2 Correlation Heatmap

Strong positive correlation between income and total spending. Catalog purchases correlate with income more than web purchases do — suggesting catalog is a premium channel. Web visits correlate negatively with spending — frequent browsers who do not convert.

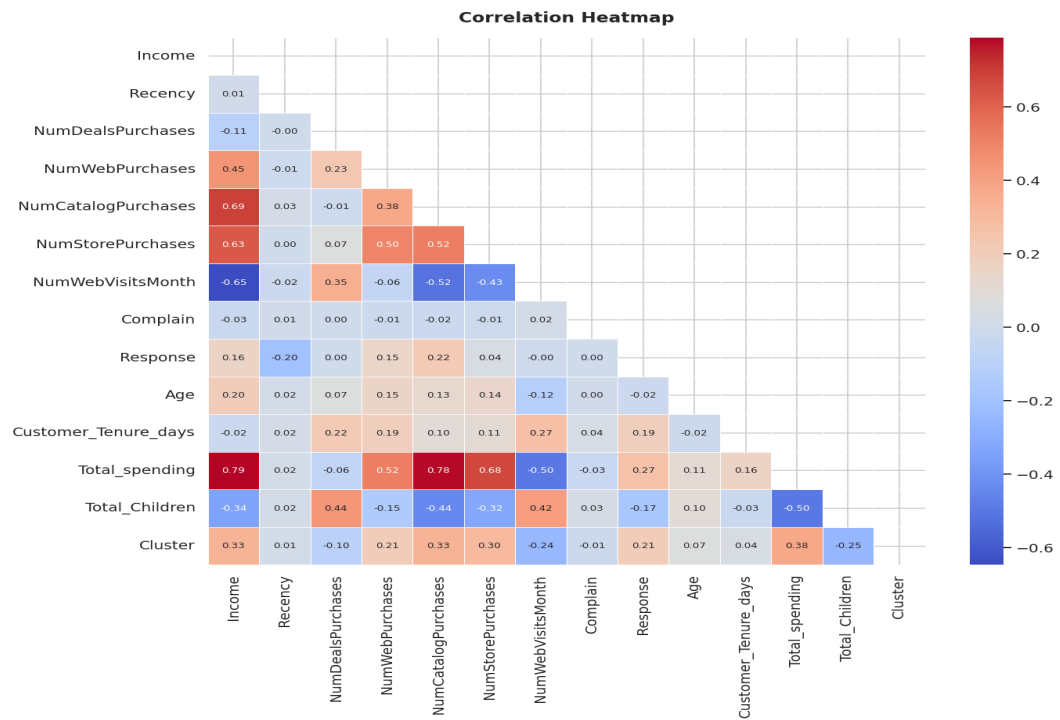


Figure 2 — Correlation heatmap (lower triangle)

5. Dimensionality Reduction — PCA

PC1 Variance	PC2 Variance	PC3 Variance	Total Explained
23.2%	11.4%	10.4%	~45%

PCA was applied to reduce the 18-feature standardised space to 3 components for visualisation and clustering. The 3 components capture approximately 45% of total variance. This is acceptable for visual exploration but means a significant portion of structure is discarded — worth noting as a limitation.

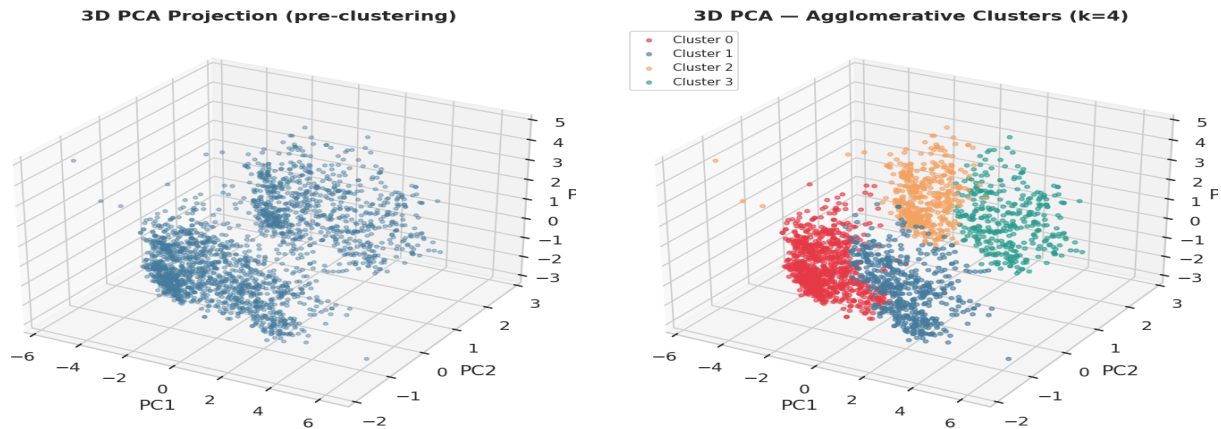


Figure 3 — Left: PCA projection pre-clustering | Right: PCA coloured by Agglomerative cluster labels

6. Optimal K Selection

Two methods were used to determine the optimal number of clusters:

- **Elbow Method (WCSS):** KneeLocator identified the elbow at $k = 4$.
- **Silhouette Score:** Computed for $k = 2$ to 10 using KMeans. Scores confirm $k = 4$ as a strong and practical choice.

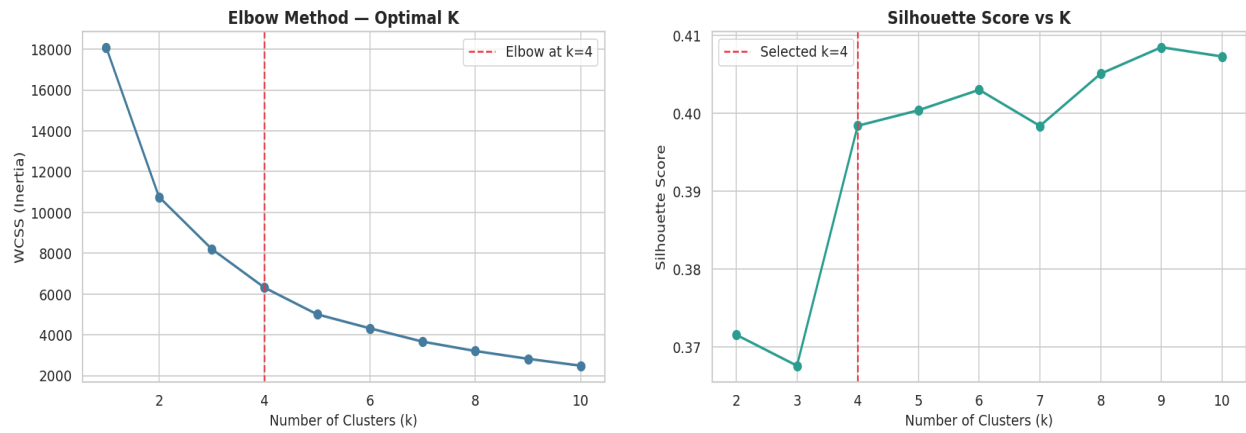


Figure 4 — Left: Elbow curve (WCSS vs k) | Right: Silhouette score vs k

7. Clustering Results

Both KMeans and Agglomerative Clustering (Ward linkage) were run at $k = 4$. Agglomerative was selected as the final model — Ward linkage tends to produce more balanced and compact clusters. Silhouette score: **0.3793** (moderate — expected for real-world customer data).

7.1 Cluster Distribution

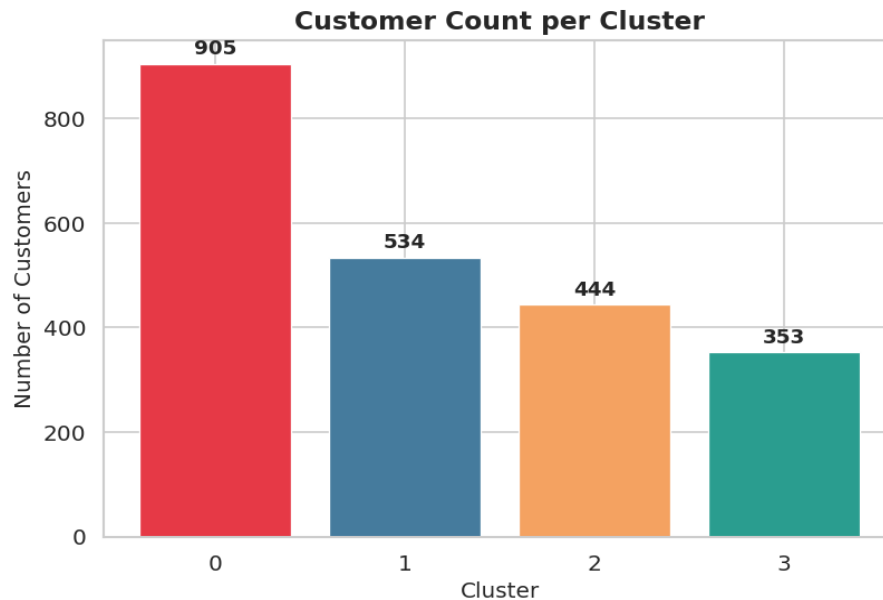


Figure 5 — Customer count per cluster

7.2 Income vs Total Spending

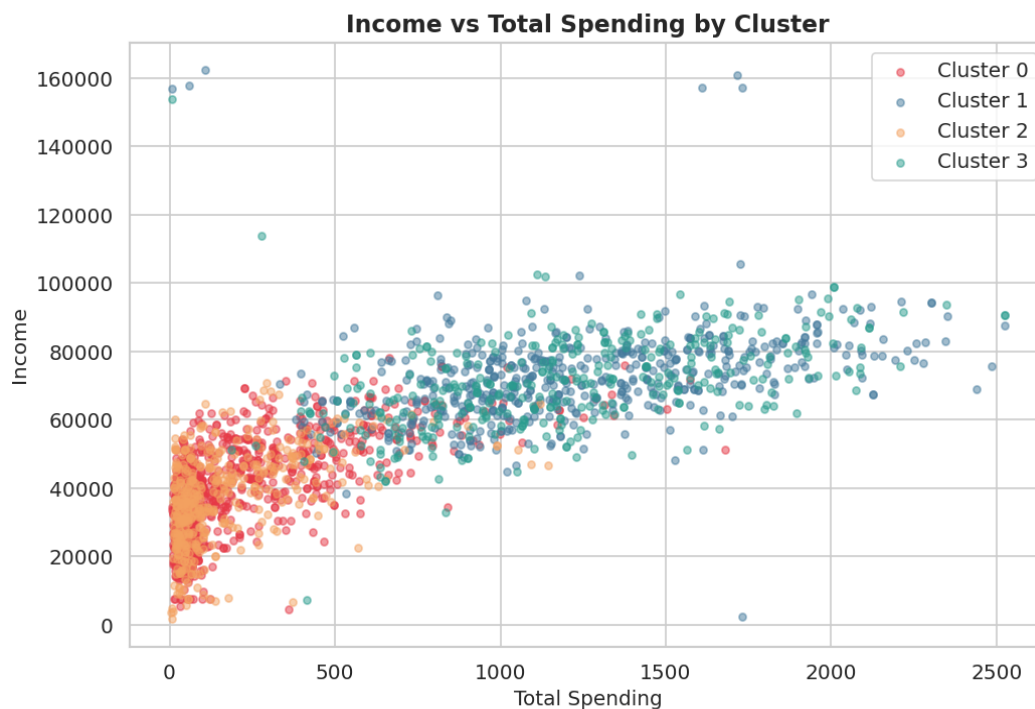


Figure 6 — Income vs Total Spending coloured by cluster. Clear separation between low-income (0, 2) and high-income (1, 3) segments.

7.3 Spending by Product Category



Figure 7 — Average spending per product category per cluster. Wines and meat are the dominant categories. High-income clusters (1, 3) outspend on every single category.

7.4 Cluster Profiles — Radar Chart

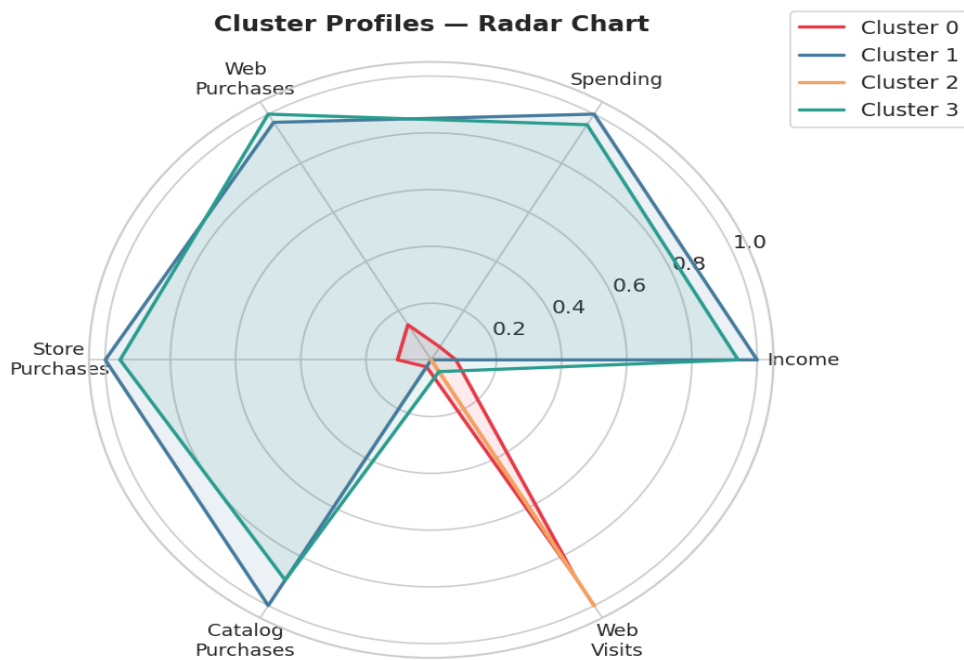


Figure 8 — Normalised radar chart of cluster profiles across key behavioural metrics. Clusters 1 and 3 dominate income and spending; Clusters 0 and 2 dominate web visits.

8. Cluster Profiles & Business Interpretation

Cluster 0	Cluster 1	Cluster 2	Cluster 3
905 customers (40%)	534 customers (24%)	444 customers (20%)	353 customers (16%)
Mid-income, moderate spending, more children, high web visits, low conversion	High income, high spending across all channels, fewer children	Lowest income and spending, highest web visits, most children	High income, high spending, standout 30% campaign response rate
Conversion campaigns, discount nudges, family bundles	Premium recommendations, loyalty rewards, early access	Budget promotions, deal alerts, essentials targeting	Direct campaign targeting, upselling, referral programs

8.1 Full Cluster Comparison Table

Metric	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Customers	905 (40%)	534 (24%)	444 (20%)	353 (16%)
Avg Income	39,681	72,808	36,960	70,723
Avg Spending	222	1,237	166	1,190
Avg Age	55.7	59.5	55.7	58.9
Avg Children	1.2	0.5	1.3	0.5
Recency (days)	48.9	49.2	48.3	50.5
Web Purchases	3.2	5.7	2.7	5.8
Store Purchases	4.1	8.7	3.6	8.4
Catalog Purchases	1.0	5.5	0.8	5.0
Web Visits/mo	6.3	3.6	6.7	3.7
Campaign Response	10%	20%	10%	30%

9. Key Findings

Income drives spending: High-income clusters (1, 3) spend 5-7 times more than low-income clusters (0, 2). Income is the single strongest predictor of spending.

Children reduce spending: Clusters with more children spend consistently less — likely constrained by family expenses regardless of income level.

High web visits do not equal high purchases: Clusters 0 and 2 visit the website most but buy least — this is a conversion problem, not a reach problem. Targeted retargeting or discount triggers could close this gap.

Recency is uniform across all clusters (~49 days): Engagement recency does not differentiate the segments. Spending behaviour is what separates customers.

Catalog purchases signal premium buyers: Near-zero for Clusters 0 and 2, high for Clusters 1 and 3. Catalog is a reliable proxy for high-value customers.

Cluster 3 is the highest-ROI campaign target: 30% response rate vs 10-20% for other clusters. Smallest group but best conversion — prioritise first.

10. Limitations

Low PCA variance retained (45%): Clustering in 3-component PCA space discards 55% of the original variance. Segments found here are approximations.

Moderate silhouette score (0.38): Clusters are real but not sharply defined — expected for behavioural data with continuous distributions.

Recency does not segment: Cannot identify dormant vs active users from cluster membership alone. A separate recency analysis or RFM model is needed.

Agglomerative does not scale: For a production system with millions of customers, KMeans or HDBSCAN is a more practical choice.

11. Recommendations

Immediate: Build separate email/notification campaigns per cluster. Target Cluster 3 first (highest campaign ROI). Run A/B tests on Clusters 0 and 2 with discount-led offers to test conversion lift.

Model improvements: Cluster on full feature space (skip PCA or use more components). Add HDBSCAN for outlier-aware clustering. Implement RFM scoring as a complementary layer.

Data improvements: Collect product-level data per channel. Track full campaign history per customer (not just binary response). Add session-level web data — visit count alone cannot explain the conversion gap in Clusters 0 and 2.